Study Session Notes 01/30/2026

Odd: $-x(t) = x(-t)$

Even: $x(t) = x(-t)$

Question 6 — 7 pts

$x(t) = e^{-t}$ $\quad x_e(t)$ $\quad x_o(t)$

x(t)=e−t can be synthesized from odd symmetric component xo(t)=_____ and even symmetric component xe(t)=_____.

odd · even

- ● xo(t)=(e−t−et)/2,xe(t)=(e−t+et)/2
- ○ xo(t)=e−t+et,xe(t)=e−t−et
- ○ xo(t)=(e−t+et)/2,xe(t)=(e−t−et)/2
- ○ xo(t)=e−t+et,xe(t)=e−t−et

1. $x_o(t) = \dfrac{e^{-t} - e^{t}}{2}, \quad x_e(t) = \dfrac{e^{-t} + e^{t}}{2}$
2. $x_o(t) = e^{-t} + e^{t}, \quad x_e(t) = e^{-t} - e^{t}$
3. $x_o(t) = \dfrac{e^{-t} + e^{t}}{2}, \quad x_e(t) = \dfrac{e^{-t} - e^{t}}{2}$
4. $x_o(t) = e^{-t} + e^{t}, \quad x_e(t) = e^{-t} - e^{t}$

1. $x_e(t) = \frac{1}{2}[x(t) + x(-t)]$

$= \frac{1}{2}[e^{-t} + e^{t}] = \dfrac{e^{-t} + e^{t}}{2}$ ✓

$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$

$= \frac{1}{2}[e^{-t} - e^{t}] = \dfrac{e^{-t} - e^{t}}{2}$ ✓

Table 1.3.1: Signal transformations.

Transformation	Expression	Consequence
Time shift	$y(t) = x(t-T)$	Waveform is shifted along $+t$ direction if $T > 0$ and along $-t$ direction if $T < 0$.
Time scaling	$y(t) = x(at)$	Waveform is compressed if $\lvert a\rvert > 1$ and expanded if $\lvert a\rvert < 1$.
Time reversal	$y(t) = x(-t)$	Waveform is mirror-imaged relative to vertical axis.
Generalized	$y(t) = x(at-b)$	Scale by a, then time shift by $\frac{b}{a}$, time reversal if $a < 0$.
Even / odd synthesis	$x(t) = x_e(t) + x_o(t)$	$x_e(t) = \frac{1}{2}[x(t) + x(-t)], \quad x_o(t) = \frac{1}{2}[x(t) - x(-t)]$

Feedback?

Question 7 — 6 pts

y(t)=x(t)cos(t) is odd symmetric if x(t) = _____

- ○ ~~t~~
- ○ ~~t2~~ t^2
- ○ ~~1+t2~~ $1+t^2$
- ● t3 t^3

1. $y(t) = (1+t)\cos(t)$ =? No!

$y(-t) = (1-t)\cos(-t)$

2. $y(t) = t^2 \cos(t)$ $\quad \cos(t) \Rightarrow$ even

$=$ even $\times$ even $=$ even

Question 8

6 pts

x(t)=2cos(4πt+π/3) is _____.

- ○ ~~not periodic~~
- ● periodic with period T0=0.5
- ○ periodic with period T0=4π
- ○ periodic with period T0=π/3

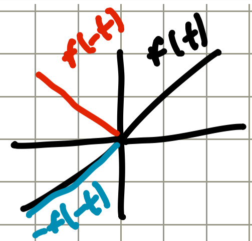
$T_0 = \frac{2\pi}{\omega} \Rightarrow \frac{2\pi}{4\pi} = 0.5$

Question 9

8 pts

x(t)=2tcos(4πt) is _____.

- ○ periodic with period T0=0.5
- ○ periodic with period T0=4π
- ● not periodic
- ○ periodic with period T0=π/2



Line

Question 10

6 pts

t2δ(t+2)= _____

$t^2 \delta(t+2)$

- ○ -4 δ(t+2)
- ○ -4
- ○ 4
- ● 4 δ(t+2)

Sampling property of $\delta(t)$

As was noted earlier, multiplying an impulse function by a constant k gives a scaled impulse of area k. Now we consider what happen when a continuous-time function $x(t)$ is multiplied by $\delta(t)$. Since $\delta(t)$ is zero everywhere except at $t = 0$, it follows that

$$x(t)\delta(t) = x(0)\delta(t),$$

provided that $x(t)$ is continuous at $t = 0$. By extension, multiplication of $x(t)$ by the time-shifted impulse function $\delta(t-T)$ gives

$$x(t)\delta(t-T) = x(T)\delta(t-T).$$

$T = -2$

$x(t)\,\delta(t-T) = x(T)\,\delta(t-T)$

$x(T) = (-2)^2 = 4$

$t^2\delta(t+2) = 4\delta(t+2)$

Question 11 — 8 pts

$\int -5 \, t2\delta(t+2)dt=$ _____

- ○ -4
- ● 4
- ○ $-4\delta(t+2)$
- ○ $4\delta(t+2)$

Multiplication of a time-continuous function $x(t)$ by an impulse located at $t = T$ generates a scaled impulse of magnitude $x(T)$ at $t = T$, provided $x(t)$ is continuous at $t = T$.

One of the most useful features of the impulse function is its ***sampling (or sifting) property***. For any function $x(t)$ known to be continuous at $t = T$:

$$\int_{-\infty}^{\infty} x(t)\delta(t-T)\,dt = x(T) \qquad \text{(sampling property)}$$

Derivation of the sampling property relies on $x(t)\delta(t-T) = x(T)\delta(t-T)$ and the unit-area property of the impulse:

$$\int_{-\infty}^{\infty} x(t)\delta(t-T)\,dt = \int_{-\infty}^{\infty} x(T)\delta(t-T)\,dt$$

$$= x(T)\int_{-\infty}^{\infty} \delta(t-T)\,dt = x(T) \times 1 = x(T).$$

$T = -2$

$$\int_{-\infty}^{\infty} x(t)\,\delta(t-T)\,dt = x(T)$$

$$\int_{-5}^{5} t^2\,\delta(t+2)\,dt \qquad T \Rightarrow_{x} \int^{y}$$

= b/c limits are -5 and 5, that determines the range that T can be in.

$x(T) = (-2)^2 = 4$

Question 12 — 7 pts

$\int -55t3\delta(-4t+8)dt=$ _____

- ○ $8\delta(-4t+8)$
- ○ -8
- ● 8
- ○ $2\delta(-4t+8)$

$$\int_{-5}^{5} t^3\,\delta(-4t+8)\,dt \Rightarrow \int_{-5}^{5} t^3\,\delta(-4(t-2))\,dt \qquad T=2$$

$+2$

$x(T) = (2)^3 = 8$

Question 13 — 7 pts

If y(t)=e−t−e−tu(t), then y(−1)= _____

$y(t) = (e^{-t} - e^{-t})u(t), \quad y(-1)$

$e^{-t} - e^{-t}u(t) \rightarrow 0$

$= e^{1} - e^{-(-1)}\cancel{u(-1)}$

- ○ 0
- ○ −e
- ● e
- ○ e(1−u(t))

Question 14 7 pts

if y(t)=r(t)−r(t−2), then y(3)= _____

- ○ -1
- ○ 0
- ● 2
- ○ 1

$y(3) = r(3) - r(3-2)$

$= 3 - 1 = 2$

$r(t) = \begin{cases} 0 & t \le 0, \\ t & t \ge 0 \end{cases}$

$r(t-2) = \begin{cases} 0 & t \le 0, \\ t-2 & t \ge 0 \end{cases}$

Question 15 8 pts

The impulse function 4δ(t−2) has height _____ and area _____.

- ○ height 1, area 4
- ○ height 4, area 2
- ○ height 4, undefined area
- ● undefined height, and area 4

δ function has area 1,
so $4\,\delta(t-2) \Rightarrow$ Area 4
Height is infinite, so height is undefined.